One-Hot Quantum Annealing for Sudoku

A QUBO Formulation of Sudoku as Graph Coloring • Benchmarked Against Classical Simulated Annealing • D-Wave Advantage2 QPU Validation

Jonah Minkoff

Dept. of Computer Science · Virginia Commonwealth University · minkoffjg@vcu.edu

VCU

COMPUTER SCIENCE

Virginia Commonwealth University

INTRODUCTION & MOTIVATION

What is Sudoku?

Sudoku is a constraint satisfaction problem (CSP) modeled as graph coloring on an N×N grid: each cell receives exactly one digit 1–N such that no digit repeats in any row, column, or √N×√N box. It encodes the same constraint structure found in scheduling, resource allocation, and combinatorial optimization.

Why Quantum Annealing?

Classical solvers (backtracking, constraint propagation, dancing links) solve 9×9 in milliseconds but scale poorly as constraints densify. Quantum annealing leverages quantum tunneling to escape local energy minima — ideal for constraint-heavy landscapes where classical methods get trapped.

Goal

Derive a QUBO formulation for Sudoku from first principles, validate correctness via simulated annealing, then test on D-Wave's Leap Hybrid BQM Solver and Advantage2 QPU. Benchmark scalability from 4×4 to 9×9, comparing quantum and classical methods on success rate and energy.

QUBO FORMULATION

Binary Variable: $x_{i,j,k} = 1$ if cell (i,j) holds digit k+1, else 0

4×4: 64 vars | 9×9: 729 vars | 9×9 with 30 givens: 459 free vars (37% reduction via given-cell elimination)

Energy Function — minimize to solve:

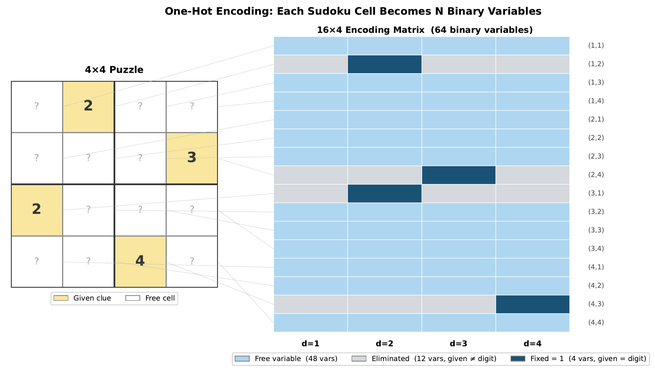
$$E_{total} = E_1 + E_2 + E_3 + E_4 = 0 \Rightarrow \text{Valid Sudoku}$$

E_1 Cell	$\Sigma_{ij} (\Sigma_k x_{ijk} - 1)^2$	— *each cell holds exactly one digit*
E_2 Row	$\Sigma_{ik} (\Sigma_j x_{ijk} - 1)^2$	— *each digit appears once per row*
E_3 Column	$\Sigma_{jk} (\Sigma_i x_{ijk} - 1)^2$	— *each digit appears once per column*
E_4 Box	$\Sigma B_o{}^x{}_{,k} (\Sigma_{(i,j)\in B_o{}^x} x_{ijk} - 1)^2$	— *each digit appears once per √N×√N box*

Matrix entries: diagonal −1 · off-diagonal (conflicting vars) +2 · **4×4:** 64×64 Q · **9×9:** 459×459 Q (30 given cells eliminated = 270 vars) · ≤28 non-zero off-diagonals per variable

$\lambda_1=\lambda_2=\lambda_3=\lambda_4=1.0$ (Lagrange multipliers, all equal — penalty weight tuning is a future direction)

ONE-HOT ENCODING



METHODOLOGY

Solvers Evaluated

- Simulated Annealing (D-Wave neal) 100 reads, 1000 sweeps
- D-Wave Leap Hybrid BQM v2.2
- D-Wave Advantage2 QPU 1,000 reads · 200 µs · forward annealing
- QAOA (Qiskit) — 2×2 only; circuit depth prohibitive at 4×4 scale — excluded from results (p=2, 2048 shots, COBYLA)

Test Instances

- 4×4 (no givens, 64 vars) — QUBO correctness baseline
- 9×9 Easy→Hard: 30/25/17 givens → 459/504/576 free vars (17 = min unique)

D-Wave Advantage2

SUDOKU TEST PUZZLES (9×9 EASY & HARD)

Sudoku Test Puzzles Used iints

9×9 Easy (30 givens) 459 free variables

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

30 givens → 459 free variables

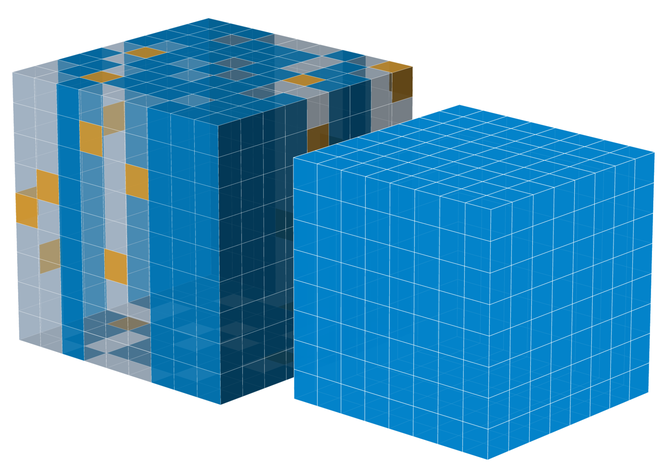
9×9 Hard (17 givens) 576 free variables

							1	
4								
	2							
				5		4		7
		8				3		
		1		9				
3			4			2		
	5		1					
			8		6			

17 givens → 576 free variables

VARIABLE REDUCTION VIA GIVEN-CELL ELIMINATION

QUBO Variable Reduction via Given-Cell Elimination
9×9 Easy Sudoku • 30 givens fixed • 270 of 729 variables eliminated



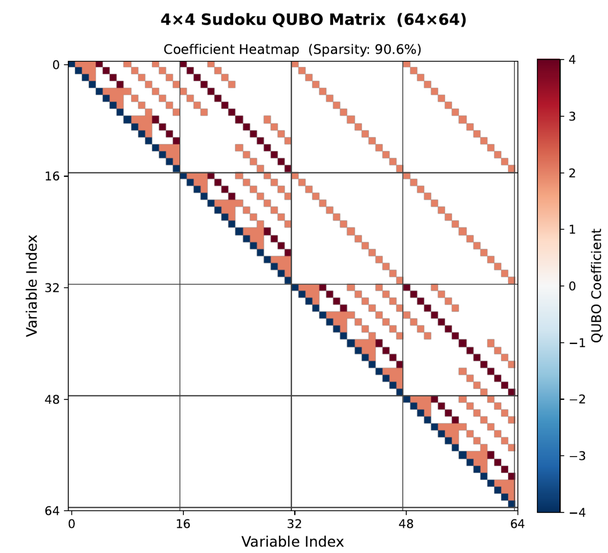
Free variable (459) · Given — correct digit (30) · Eliminated — wrong digit (240)

RESULTS: D-WAVE QPU VS. SIMULATED ANNEALING (4×4 Sudoku)

D-Wave Advantage2 QPU vs. Simulated Annealing — 4×4 Sudoku
Forward Annealing • 1,000 reads • 200 µs • 4 puzzles per difficulty

Difficulty	Givens	Free Vars	Qubits	SA Opt %	QPU Opt %	QPU ±Std
Easy	12	16	16–17	100.0%	100.0%	±0.0%
Medium	8	32	46–47	100.0%	99.4%	±1.0%
Hard	4	48	99–117	100.0%	76.1%	±12.7%

QUBO COEFFICIENT MATRIX (4×4 SUDOKU, 64×64)



KEY OBSERVATIONS

- SA degrades sharply with scale: 99% on blank 4×4 (64 vars) → 7% on 9×9 Easy (459 vars, 30 givens) — 7× more variables, 14× lower success. Note: baselines differ (blank 4×4 vs. given-reduced 9×9); SA's collapse is likely understated on a fully blank 9×9.
- QPU on 4×4: 100% Easy · 99.4% Medium · 76.1% ± 12.7% Hard — confirms the QUBO formulation produces valid Sudoku solutions on real quantum hardware.
- SA outperforms QPU on 4×4 Hard (100% vs. 76.1%) — QPU struggles on the densest constraint instances even at small scale.
- Chain breaks: 0% on all 12 QPU experiments — the Pegasus topology embeds 4×4 Sudoku cleanly; Hard failures are an annealing quality issue, not a connectivity issue.
- QPU access time: 337–352 ms per experiment — hardware overhead is modest; bottleneck is solution quality, not speed.

CONCLUSIONS

- QUBO encoding is provably exact: E = 0 iff all four constraint types hold — validated on both 4×4 and 9×9.
- SA drops from 99% (4×4 blank) to 7% (9×9 Easy) with a 7× variable increase — classical annealing does not scale.
- QPU finds valid Sudoku on real hardware: 100% Easy, 99.4% Medium, 76.1% Hard on 4×4 instances.
- SA beats QPU on 4×4 Hard (100% vs. 76.1%) despite 0% chain breaks — energy landscape complexity is the bottleneck, not embedding.
- Direct 9×9 QPU benchmarking remains open — 4×4 Hard results motivate annealing schedule and encoding improvements first.

FUTURE DIRECTIONS

- Per-constraint penalty tuning ($\lambda_1 \ldots \lambda_4$) to improve Hard-instance success rate
- Alternative encoding (e.g. domain-wall) + embedding optimization to cut qubit count and chain breaks
- Direct Advantage2 QPU on full 9×9 without hybrid — pure annealing benchmark
- Sweep full clue-count range (17–40) to map where the QPU's solvability boundary lies
- Scale to 16×16 Sudoku via hybrid quantum-classical decomposition

Special Thanks: Prof. Thang Dinh, Dept. of Computer Science, Virginia Commonwealth University — for guidance, resources, and D-Wave hardware access.

RESOURCES